

Chapter 2: Prompt Engineering & AI Content Creation

Class 8: Practical / Activities (10 Minutes)

Topic: Generate a Research Report Using AI

Activity 1: Generate a Research Report Using AI (4 Minutes)

Objective

To help students use AI tools to generate a well-structured research report by writing effective prompts, collecting relevant information, organizing content under proper headings, and understanding the importance of reviewing AI-generated reports.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, or Perplexity AI)
 - Student Research Report Worksheet
 - Notebook & Pen
 - Teacher Observation Sheet
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Instructions

1. Open any AI tool.
2. Select one research topic from the list below.
3. Write a detailed research prompt.
4. Generate a research report using AI.
5. Read the report carefully.

6. Check whether it includes:
 - Title
 - Introduction
 - Objectives
 - Main Content
 - Findings
 - Conclusion
 7. Organize the report neatly in the worksheet.
 8. Review and improve the report before submission.
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Suggested Research Topics

- Artificial Intelligence
 - Climate Change
 - Renewable Energy
 - Cyber Security
 - Water Conservation
 - Plastic Pollution
 - Digital Citizenship
 - Space Exploration
 - Healthy Lifestyle
 - Sustainable Development
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Sample AI Prompt

Prompt:

"Act as a research expert. Prepare a research report on Artificial Intelligence for Class X students with a title, introduction, objectives, key findings, advantages, disadvantages, conclusion, and references in simple English."

Student Research Report Worksheet

Task	Completed (✓)
Topic Selected	☐
AI Prompt Written	☐
Research Report Generated	☐

Task	Completed (✓)
Report Organized	☐
Report Reviewed	☐
Final Report Completed	☐

Student Activity Checklist

Students should:

- ✓ Select an appropriate topic.
 - ✓ Write a detailed AI prompt.
 - ✓ Generate a structured research report.
 - ✓ Review AI-generated content.
 - ✓ Organize the report correctly.
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Teacher Observation Checklist

- ✓ Writes an effective research prompt.
 - ✓ Generates a complete report.
 - ✓ Organizes information properly.
 - ✓ Reviews AI output carefully.
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Activity 2: Improve and Verify the Research Report (3 Minutes)

Objective

To help students evaluate AI-generated reports, verify important facts using reliable sources, and improve the quality of the final report.

Materials Required

- School Textbook or Reliable Educational Source
 - AI Research Report
 - Report Evaluation Worksheet
 - Teacher Evaluation Sheet
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Instructions

1. Read the AI-generated report.
 2. Verify important facts using:
 - School textbook
 - Teacher notes
 - Reliable educational websites
 3. Correct any inaccurate information.
 4. Improve the report by:
 - Adding missing details
 - Improving headings
 - Correcting grammar
 - Simplifying language
 5. Rewrite the final report in your own words.
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Research Report Evaluation Worksheet

Evaluation Criteria Yes Needs Improvement

Clear Title	☐	☐
Good Introduction	☐	☐
Correct Information	☐	☐
Proper Headings	☐	☐
Logical Conclusion	☐	☐
References Included	☐	☐

Discussion Questions

1. Why should AI-generated reports be verified?
2. Which section of the report was most important?
3. How did AI help organize the report?
4. What improvements did you make to the report?

Teacher Observation Checklist

- ✓ Verifies facts carefully.
 - ✓ Identifies errors.
 - ✓ Improves report quality.
 - ✓ Demonstrates critical thinking.
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Activity 3: Present the Research Report & Reflection (3 Minutes)

Objective

To help students present their AI-generated research report, explain the prompt used, and reflect on the benefits and limitations of AI-assisted report writing.

Materials Required

- Final Research Report
 - Reflection Worksheet
 - Teacher Observation Sheet
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Instructions

1. Present your report in **1 minute**.
2. Explain:
 - Research topic
 - AI prompt used
 - Three important findings
 - One improvement made after review
3. Complete the reflection worksheet.
4. Share one responsible AI practice you followed.

Reflection Worksheet

Answer the following questions:

1. What topic did you research?

2. Write the AI prompt you used.

3. What were the three main findings of your report?

1.

2.

3.

4. What changes did you make after reviewing the AI report?

5. Why is verifying AI-generated information important?

Self-Assessment Checklist

Skill	Yes	Needs Practice
I wrote an effective research prompt.	☐	☐
I generated a structured report.	☐	☐
I verified important information.	☐	☐
I improved the report quality.	☐	☐
I presented my report confidently.	☐	☐

Teacher Observation Checklist

- ✓ Presents the report clearly.
 - ✓ Explains prompt usage effectively.
 - ✓ Demonstrates responsible AI practices.
 - ✓ Participates actively in the activity.
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Mini Practical Assessment Rubric

Assessment Area	Marks
Research Prompt Writing	20
AI Research Report Generation	20
Report Organization & Formatting	20
Report Verification & Improvement	20
Presentation, Reflection & Participation	20
Total	100 Marks

Teacher Observation Checklist

- ✓ Uses AI tools effectively to generate a structured research report.
 - ✓ Writes clear and detailed prompts for research.
 - ✓ Verifies AI-generated information using reliable educational sources.
 - ✓ Organizes and presents the research report professionally.
 - ✓ Demonstrates responsible AI use, critical thinking, and active participation throughout the practical activity.
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Practical Time Distribution

Activity	Duration
Generate a Research Report Using AI	4 Minutes
Improve & Verify the Research Report	3 Minutes
Present the Report & Reflection	3 Minutes
Total Practical Time	10 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Write **effective AI prompts** to generate well-structured research reports.
- Use AI tools to create reports with **titles, introductions, objectives, findings, conclusions, and references**.
- Verify and improve AI-generated reports using reliable educational sources.
- Organize research content into a professional report format suitable for academic projects.
- Present research findings confidently while explaining the AI prompt used.
- Demonstrate responsible, ethical, and accurate use of AI for **research, report writing, assignments, and project work**.